9 A satellite of mass 6.9×10^2 kg orbits the Earth in a circular orbit of radius 6.5×10^6 m. Its orbital speed is 7.9 km s^{-1}.

A rocket engine is fired such that the satellite moves into a higher orbit of radius 7.2×10^6 m where its orbital speed decreases to 7.5 km s^{-1}.

Which statement about the change in total energy of the satellite is correct?

Mass of the Earth = 6.0×10^{24} kg

A The satellite gains approximately 2×10^9 J.

B The satellite gains approximately 4×10^9 J.

C The satellite loses approximately 2×10^9 J.

D The satellite loses approximately 6×10^9 J.